Workflow Overview

A typical data science project follows these steps: define the problem, collect data, clean the data, explore the data, engineer features, train models, evaluate them and deploy the best one. The steps are repeated many times rather than done once.

Data Cleaning and Missing Values

Real data is messy. Common problems are duplicate rows, wrong data types, outliers and missing values. Missing values can be handled by dropping rows, filling with the mean or median, or using a model-based imputer. The right choice depends on how much data is missing and why.

Feature engineering is the process of creating new input columns that help a model learn, for example extracting the month from a date or computing the ratio of two columns. Good features often matter more than the choice of algorithm.

Model Evaluation

Data is split into training, validation and test sets. The model learns from the training set, the validation set is used to tune settings, and the test set is used once at the end to estimate real-world performance. Overfitting happens when a model scores well on training data but poorly on new data. Cross-validation repeats the split several times to give a more reliable estimate.